

Data Construction Guide

Monthly Stock Return Prediction Project

A concise guide to the five-file data pipeline and the final model-ready dataset

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1 Objective and Final Dataset

The pipeline constructs a monthly stock-level dataset for predicting next-month excess returns. Each observation is one ticker-month. Predictors are measured using information available by the end of month t , and the target is the stock’s excess return in month $t + 1$:

$$y_{i,t+1} = r_{i,t+1} - RF_{t+1}.$$

The final model-ready file is intended to be `output/data/final/model_dataset_kelly_ranked_full_1990_2025.parquet`. It contains `ticker`, `month`, the raw target `target_excess_return_next_1m`, and rank-normalized predictors. The common data file does not define train, validation, or test samples; chronological splitting is deferred to the model workflow.

The sample starts in January 1990 and ends in December 2025. Daily price data begins in 1987 so that long-window predictors can be formed by 1990. The price window ends in early 2026 because January 2026 returns are needed to realize the December 2025 next-month target.

2 Backward Construction Map

Stage		Main file	Role
Final panel	ranked	05_clean_and_rank_normalize.py	Drops observations without a target, keeps usable numeric predictors, imputes within month, and rank-normalizes predictors to $[-1, 1]$.
Raw predictor panel	predictor	04_build_raw_kelly_dataset.py	Selects stock characteristics, adds market/VIX controls, SIC2 dummies, Welch–Goyal states, and stock-characteristic $\times$ macro interactions.
Monthly panel with fundamentals	panel	03_add_fundamentals_and_macro.py	Constructs annual Compustat variables and merges only the most recent report available after a six-month lag.
Monthly stock panel	stock	02_build_monthly_stock_features.py	Converts daily prices into monthly returns, risk, liquidity, trend variables, beta, idiosyncratic volatility, and the next-month excess-return target.
Clean prices	daily	01_build_clean_yahoo_daily.py	Loads the locked stock universe, downloads daily Yahoo prices, removes bad rows, and removes ticker histories only when bad rows exceed the configured threshold.
Permanent inputs		input/	Stores locked stock universe, Compustat, Fama–French RF, market/VIX, and Welch–Goyal macro inputs.

3 Data Sources and Permanent Inputs

Normal pipeline execution reads permanent local inputs. Internet acquisition logic is kept outside the production data files in `src/acquisition/`.

Input file	Source	Use
<code>input/stock_universe_locked.csv</code>	Wikipedia S&P 500 current and historical constituents	Locked Yahoo-format stock universe. Must contain <code>ticker</code> .
<code>input/raw/compustat_annual_1980_2025.csv</code>	Compustat annual fundamentals	Annual accounting characteristics and SIC2 industry codes.
<code>input/fama_french_rf_monthly.csv</code>	Kenneth French data library	Monthly risk-free rate in decimal form; columns <code>month</code> , <code>RF</code> .
<code>input/market_gspc_daily.csv</code>	Yahoo Finance <code>^GSPC</code>	Market return, market volatility, beta, and idiosyncratic volatility.
<code>input/market_vix_daily.csv</code>	Yahoo Finance <code>^VIX</code>	Monthly VIX level and VIX change.
<code>input/external/welch_goyal_macro_1990_2025.csv</code>	Cleaned Welch–Goyal input	Eight monthly macro states used directly and in interactions.

The Welch–Goyal input currently also contains `wg_infl`, but the locked interaction design intentionally excludes it. File 04 uses only `wg_dp`, `wg_ep`, `wg_bm`, `wg_ntis`, `wg_tbl`, `wg_tms`, `wg_dfy`, and `wg_svar`, matching the intended Gu–Kelly–Xiu style eight-macro-state specification.

4 File 01: Daily Stock Prices and Quality Cleaning

Input: `input/stock_universe_locked.csv`.

Output: clean daily prices, ticker-quality report, and removed-ticker report.

Purpose: create a clean daily OHLCV stock-price panel.

File 01 loads the locked universe, downloads Yahoo data with `auto_adjust=False`, standardizes columns, and computes daily adjusted returns:

$$r_{i,d} = \frac{AdjClose_{i,d}}{AdjClose_{i,d-1}} - 1.$$

The daily cleaning rules remove bad rows for:

- impossible OHLC relations, such as `high < low`;
- missing core fields: open, high, low, close, adjusted close, or volume;
- nonpositive adjusted prices;
- duplicate ticker-date rows;
- extreme daily returns above 300% or below −95%.

A whole ticker is removed only if its bad-row share exceeds `DAILY_BAD_ROW_SHARE_THRESHOLD`, currently 10%. This avoids throwing away a complete stock history because of a small number of bad observations.

5 File 02: Monthly Stock Features and Target

Input: clean daily stock prices, permanent GSPC/VIX daily files, and permanent Fama–French RF.

Output: `monthly_stock_panel_with_targets_1990_2025.csv`.

Purpose: construct monthly price-based predictors and the next-month excess-return target.

Monthly return is computed from month-end adjusted prices:

$$r_{i,t} = \frac{AdjClose_{i,t}}{AdjClose_{i,t-1}} - 1.$$

Momentum variables are compounded from lagged monthly returns, so the current month is excluded:

$$mom_{i,t}^{(k)} = \prod_{j=1}^k (1 + r_{i,t-j}) - 1.$$

The retained momentum variables are `mom6m`, `mom12m`, and `long_term_return_36m`. All are based on lagged monthly returns, so the current month is excluded.

Liquidity and trading variables include Amihud illiquidity:

$$Amihud_{i,d} = \frac{|r_{i,d}|}{AdjClose_{i,d} \times Volume_{i,d}},$$

then averaged within month as `amihud_1m`. Other monthly variables include realized return volatility, minimum daily return, average of the five largest daily returns, zero-volume share, dollar-volume growth, price-to-moving-average, distance from the 12-month high, and average intramonth range.

The file removes ticker histories with repeated implausible monthly returns. A ticker is removed if it has at least two monthly returns above 300% or below -95%.

Market and VIX controls are added directly. VIX is used as an aggregate volatility and risk-aversion state variable following the standard Cboe VIX methodology [4]. These four variables are direct market-state predictors, not Welch–Goyal variables, and they are not interacted with the macro states:

`market_ret_1m`, `market_vol_1m`, `vix_avg_1m`, `vix_change_1m`.

Rolling beta and idiosyncratic volatility are estimated from daily stock and market returns over a 252-trading-day window, with at least 126 observations:

$$\beta_{i,t} = \frac{\text{Cov}(r_{i,d}, r_{m,d})}{\text{Var}(r_{m,d})}, \quad \sigma_{i,t}^{idio} = \sqrt{\text{Var}(r_{i,d}) - \beta_{i,t}^2 \text{Var}(r_{m,d})}.$$

The locked risk block also includes `betasq_12m`, calculated as `beta_12m ** 2`.

The target is created by shifting each stock’s monthly return forward one month and subtracting the next-month risk-free rate:

$$\text{target_excess_return_next_1m}_{i,t} = r_{i,t+1} - RF_{t+1}.$$

6 File 03: Compustat Characteristics and Timing

Input: monthly stock panel and raw annual Compustat file.

Output: `monthly_panel_with_compustat_macro_1990_2025.csv`.

Purpose: add accounting characteristics without look-ahead bias.

Compustat `datadate` is the fiscal-year-end date, not the date at which the report is publicly available. The project therefore imposes a conservative six-month availability lag:

$$AvailableMonth = MonthEnd(datadate + 6 \text{ months}).$$

For each ticker-month, the merge uses the most recent Compustat report whose available month is not later than the stock month. This is a backward as-of merge and prevents future accounting information from entering predictors.

The retained accounting variables are compact and standard:

Family	Current variables
Valuation and size	<code>be_me</code> , <code>ocf_me</code> , <code>log_comp_market_equity</code>
Profitability	<code>op_at</code> , <code>ocf_at</code>
Leverage and liquidity	<code>debt_at</code> , <code>cash_at</code> , <code>cashflow_to_debt</code> , <code>current_ratio</code> , <code>quick_ratio</code>
Investment and asset structure	<code>capx_at</code> , <code>capx_growth</code> , <code>rd_at</code> , <code>ppe_at</code> , <code>tangibility</code> , <code>asset_turnover</code> , <code>sales_to_inventory</code> , <code>sales_to_cash</code> , <code>sales_to_receivables</code>
Accruals, payout, and financing	<code>accruals_at</code> , <code>dividend_dummy</code> , <code>equity_issuance_at</code> , <code>repurchase_at</code>
Growth and missingness	<code>asset_growth</code> , <code>sales_growth</code> , <code>ppeinv_gr1a</code> , <code>xrd_missing</code>

Key ratios include book-to-market `be_me`, operating cash flow to market equity `ocf_me`, operating profitability to assets `op_at`, debt to assets `debt_at`, cash to assets `cash_at`, cash flow to debt `cashflow_to_debt`, current and quick ratios, accruals to assets `accruals_at`, sales-scaled inventory/cash/receivables ratios, and annual growth rates for assets, sales, capital expenditures, and property-plus-inventory.

7 File 04: Raw Predictor Panel and Interactions

Input: monthly panel with Compustat variables and cleaned Welch–Goyal macro input.

Output: raw Kelly-style model dataset and raw predictor list.

Purpose: define the final raw predictor matrix before imputation and normalization.

File 04 keeps 47 locked stock characteristics and 4 direct market/VIX controls. It adds SIC2 industry dummies, including `sic2_missing`, and merges eight locked Welch–Goyal macro states. The interactions are:

$$Interaction_{i,t}^{c,m} = Characteristic_{i,t}^c \times MacroState_t^m.$$

Their purpose is to allow the predictive relevance of firm characteristics to vary with aggregate conditions, consistent with conditional expected returns and state-dependent risk premia. The code does not create market $\times$ macro, VIX $\times$ macro, macro $\times$ macro, SIC2 $\times$ macro, or target interactions.

With the locked design, file 04 creates $47 \times 8 = 376$ interaction variables. SIC2 dummy count is data-dependent because it depends on observed industries in the merged panel.

8 File 05: Imputation and Rank Normalization

Input: raw Kelly-style predictor panel and raw predictor list.

Output: ranked full-sample parquet file, ranked predictor list, and cleaning diagnostics.

Purpose: produce the final model-ready full sample.

File 05 drops observations with missing targets because supervised models require observed outcomes. It keeps numeric predictors, drops only predictors that are completely empty, and keeps the raw target rather than winsorizing it. Extreme targets are documented in diagnostic files but not removed from the common dataset.

For each month, missing predictor values are replaced by that month’s cross-sectional median. If a predictor is missing for the full month, the fallback value is zero. The predictors are then rank-normalized within month:

$$x_{i,t}^{rank} = 2 \frac{rank(x_{i,t}) - 1}{N_t - 1} - 1,$$

where N_t is the number of firms in month t . If a month has only one firm, the transformed value is set to zero. This transformation preserves cross-sectional order, reduces the influence of outliers, and puts predictors on a common scale.

9 Output Files and Diagnostics

The pipeline saves both data files and audit files. The data files are used by later model scripts; the audit files are used to verify that the construction is behaving as intended.

File type	Main files
Daily price outputs	daily_prices_raw_1987_2026.csv, daily_prices_clean_1987_2026.csv, sp500_tickers_clean.csv
Daily quality diagnostics	ticker_quality_report.csv, removed_tickers.csv
Monthly stock outputs	monthly_stock_panel_with_targets_1990_2025.csv, monthly_return_quality_report.csv, monthly_removed_tickers.csv
Fundamental merge out-puts	compustat_annual_cleaned_1980_2025.csv, monthly_panel_with_compustat_macro_1990_2025.csv
Raw model outputs	model_dataset_kelly_raw_full_1990_2025.csv, predictor_columns_kelly_raw.csv, kelly_raw_dataset_summary.csv
Final model-ready out-puts	model_dataset_kelly_ranked_full_1990_2025.parquet, predictor_columns_kelly_ranked.csv, cleaning_summary.csv
Final diagnostics	dropped_predictors_empty.csv, monthly_imputation_medians_summary.csv, extreme_target_observations.csv, extreme_target_counts.csv

The most important diagnostic is `cleaning_summary.csv`. It should confirm zero missing values, zero missing final targets, zero duplicate ticker-month rows, raw target treatment, and deferred sample splitting. The predictor-column files are also important because they freeze the feature list passed to modeling scripts.

10 Timing and Leakage Controls

Rule	Implementation
Predictor timing	Stock predictors dated month t use prices, volume, market data, VIX, macro states, and accounting information observable by month t .
Target timing	The target is month $t + 1$ stock return minus month $t + 1$ RF.
Momentum timing	Momentum and long-horizon return use lagged returns and exclude the current month.
Accounting timing	Annual Compustat reports become usable six months after fiscal-year end.
Accounting merge	Backward as-of merge uses only reports whose available month is not later than the stock month.
Preprocessing timing	Imputation and rank normalization use only the same month's cross-section.
Model splitting	Train/validation/test splits happen later in model scripts, not in the common data-construction file.

11 Final Predictor Groups and Outputs

Group	Retained variables
Recent return and momentum	ret_1m, mom6m, mom12m, long_term_return_36m
Volatility and tail risk	retvol_1m, retvol12m, minret_1m, rmax5_21d, beta_12m, betasq_12m, idiovol_12m
Liquidity and trading activity	avg_dolvol_1m, std_dolvol_1m, avg_log_dolvol_1m, amihud_1m, zerotrade_1m, dolvol_growth_1m
Price trend	price_to_ma12, dist_from_high_12m, avg_range_1m
Market and VIX state	market_ret_1m, market_vol_1m, vix_avg_1m, vix_change_1m
Valuation and size	be_me, ocf_me, log_comp_market_equity
Profitability	op_at, ocf_at
Leverage and liquidity	debt_at, cash_at, cashflow_to_debt, current_ratio, quick_ratio
Investment and asset structure	capx_at, capx_growth, rd_at, ppe_at, tangibility, asset_turnover, sales_to_inventory, sales_to_cash, sales_to_receivables
Accruals, payout, and financing	accruals_at, dividend_dummy, equity_issuance_at, repurchase_at
Growth	asset_growth, sales_growth, ppeinv_gr1a, xrd_missing
Industry controls	SIC2 dummies, including sic2_missing
Macro states	wg_dp, wg_ep, wg_bm, wg_ntis, wg_tbl, wg_tms, wg_dfy, wg_svar
Interactions	each retained stock characteristic $\times$ each retained Welch–Goyal macro state

The main literature motivation is compact: momentum variables follow the return-predictability literature [9]; price-trend variables follow the 52-week-high logic of George and Hwang [6]; valuation and profitability variables follow cross-sectional asset-pricing evidence [5, 10]; volatility, downside, and maximum-return variables follow risk and lottery-demand evidence [2, 3]; illiquidity follows Amihud [1]; accounting accruals and investment variables follow anomaly evidence from Sloan and Titman, Wei, and Xie [11, 12]; macro states follow Welch and Goyal [13]; interactions are motivated by modern machine-learning evidence in Gu, Kelly, and Xiu [7]. Some retained variables reproduce Kelly-style characteristics closely, while others are Yahoo or annual-Compustat proxies. Many Kelly variables remain unavailable because they require CRSP microstructure fields, quarterly Compustat, announcement dates, analyst data, or other external inputs.

12 Current Statistics and Verification Status

After rerunning files 02–05 with the existing local inputs, the verified final dataset statistics are:

- final rows: 212,801;
- final columns: 494;
- final predictors: 491;
- unique tickers: 663;
- first month: 1990-01-31;
- last month: 2025-12-31;
- missing targets: 0;

- missing values: 0;
- duplicate ticker-month rows: 0;
- locked stock characteristics: 47;
- direct market/VIX variables: 4;
- locked Welch–Goyal macro states: 8;
- characteristic $\times$ macro interactions: 376;
- SIC2 dummies: 64.

The final ranked predictors lie in $[-1, 1]$, all locked characteristics are present, all locked interactions are present, and final predictor names are unique. The Kelly-characteristic comparison report is saved as `output/reports/kelly_characteristic_comparison.csv`.

13 Validation Checklist

After rerunning the pipeline, the following checks should hold before using the file in model estimation:

- the final panel has one row per ticker-month and zero duplicate ticker-month rows;
- the final target column has no missing values;
- all retained predictors are numeric;
- no predictor is completely empty after the target drop;
- monthly median imputation leaves no missing predictor values;
- every rank-normalized predictor lies inside $[-1, 1]$;
- the ranked predictor list matches the predictor columns in the final parquet file;
- the final month is December 2025 and the first model month is January 1990;
- the raw target remains unwinsorized in the common data file;
- any model-specific winsorization or scaling is fitted only inside the model workflow.

These checks are intentionally mechanical. They do not prove that the data is economically perfect, but they verify the central construction contract: correct keys, complete predictors, preserved raw target, chronological timing, and no accidental data leakage in the common panel.

14 Reproduction Steps

From the project root, run:

1. `python src/data/01_build_clean_yahoo_daily.py`
2. `python src/data/02_build_monthly_stock_features.py`
3. `python src/data/03_add_fundamentals_and_macro.py`
4. `python src/data/04_build_raw_kelly_dataset.py`
5. `python src/data/05_clean_and_rank_normalize.py`

The normal pipeline should not download Wikipedia, Fama–French, GSPC, VIX, or Welch–Goyal inputs. If those inputs need to be rebuilt, use the one-time scripts in `src/acquisition/`.

15 Limitations

The project uses a locked S&P 500-style universe rather than a full CRSP universe. This is practical and transparent but does not fully remove survivorship or selection concerns. Yahoo tickers are also less stable than permanent security identifiers, so the cleaning reports are important. Annual Compustat data is lagged conservatively, but exact filing dates are not used. Finally, the compact predictor set is designed for a clean academic machine-learning project; it is not a full replication of every published anomaly variable.

16 References

References

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